NATIONAL INSTISTUTE OF BUSINESS MANAGEMENT

SCHOOL OF COMPUTING & ENGINEERING



HIGHER DIPLOMA IN SOFTWARE ENGINEERING

26.1F

Statistics for Computing

The Impact of Algorithmic Social Media on Digital Media Addiction

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Abstract

With the advent of digital-media architectures, users' engagement has moved from voluntary and pull-based engagement to involuntary and push-based engagement facilitated by algorithms. The research paper focuses on predicting the algorithmic mechanisms in relation to digital media addiction amongst the youth population in Sri Lanka. Underpinned by Operant Conditioning Theory and Media System Dependency Theory, the study focuses on analyzing four primary sociotechnical constructs: Algorithmic Content Recommendations, Parasocial Influencer Attachment, Echo Chamber Effect, and Cultural Content Shift. Utilizing positivism as epistemology and deductive approach as the research design, the data was gathered using a structured questionnaire based on Likert scale which was administered using the Google OAuth form for authenticating respondents [cite: 17, 20]. The data analysis involved the use of Pearson Chi-Square, One-Way ANOVA, Pearson Bivariate Correlation, and OLS Multiple Linear Regression tests.

Empirical results demonstrate extremely significant total variance of 51.8, resulting from four independent variables related to digital media addiction (R-squared = 0.518, $F(4, 317) = 85.009$, $p < 0.001$) [cite: 17, 20]. As part of the discussed conceptual framework, recommendation algorithms are considered to be the key factor of influence ($\beta = 0.384$, $p < 0.001$), along with parasocial attachment to influencers ($\beta = 0.285$, $p < 0.001$) [cite: 17, 20]. Demographical studies were carried out using the Chi-Square test; thus, a statistically significant choice of preferred platforms according to gender ($p = 0.042$), as well as high levels of digital addiction among all adults (ANOVA, $p = 0.052$) were established [cite: 17, 20]. It means that digital media addiction results from the mathematically predicted consequences of modern dark patterns and recommendation mechanisms in UI/UX design, rather than lack of self-control on the part of users.

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Chapter 1: Introduction and Conceptual Framework

1.1 Background of the Study: The Algorithmic Paradigm Shift

There have been many changes in the structure of digital media in the last ten years; it has evolved from the decentralized model of Web 1.0 and Web 2.0 to an algorithmic curation model that we see in today's digital platform ecosystem. Digital media consumption was first driven by a "pull" model, whereby consumers went to certain URLs or made specific searches to access the desired information. The current global platforms such as TikTok, YouTube, Instagram, and Facebook work on a "push" model. The current research uses the recent development in recommender systems based on the use of deep neural networks, matrix factorization, and telemetry in order to process the large amount of implicit user data that includes, for example, the time a user stays on the certain screen and the speed of his/her swipes and taps. This way, vector embeddings of user's behavior are created and the algorithms for filtering out personalized content streams are built. Even though the above-described method helps to find new content, its software engineering goal is purely mathematical and is related to user retention.

1.2 Problem Statement

Frictionless interfaces, which include infinite scrolling, automatic feed refreshes, "Up Next" playlists, and algorithmic push notifications, have led to various unexpected sociotechnical consequences. Specifically, young adults who are involved in the academic or technological fields often exhibit behaviors that are characterized by excessive use of social media, which is linked to poor academic achievement, low-quality sleep, and increased levels of distraction. Though fatigue from the use of computers through screens is generally known by the population, there is no scientific proof that shows any statistical correlation between specific software designs in algorithmic choices and the levels of addiction to the use of computers. Without such a scientific basis, engineers have no reason to adopt such measures.

1.3 Research Objectives

Objective of the research: development and assessment of a predictive model measuring the impact of algorithms and the structure of social media on digital media addiction among youth.

- Objective 1: To measure the impact of algorithms for distributing content (autoplay, playlist, push notifications) on users' addiction to the application.
- Objective 2: To explore the role of parasocial interaction with influencers in addictive behavior in the application.

- Objective 3: To measure the impact of creating an algorithmic echo chamber on cognitive absorption and extensive scrolling of content.
- Objective 4: To measure the impact of culturally related changes caused by algorithms, namely, the shift from local to foreign content on the platform addiction.
- Objective 5: To predict the variables above using the OLS regression model.

1.4 Conceptual Framework and Structural Modeling

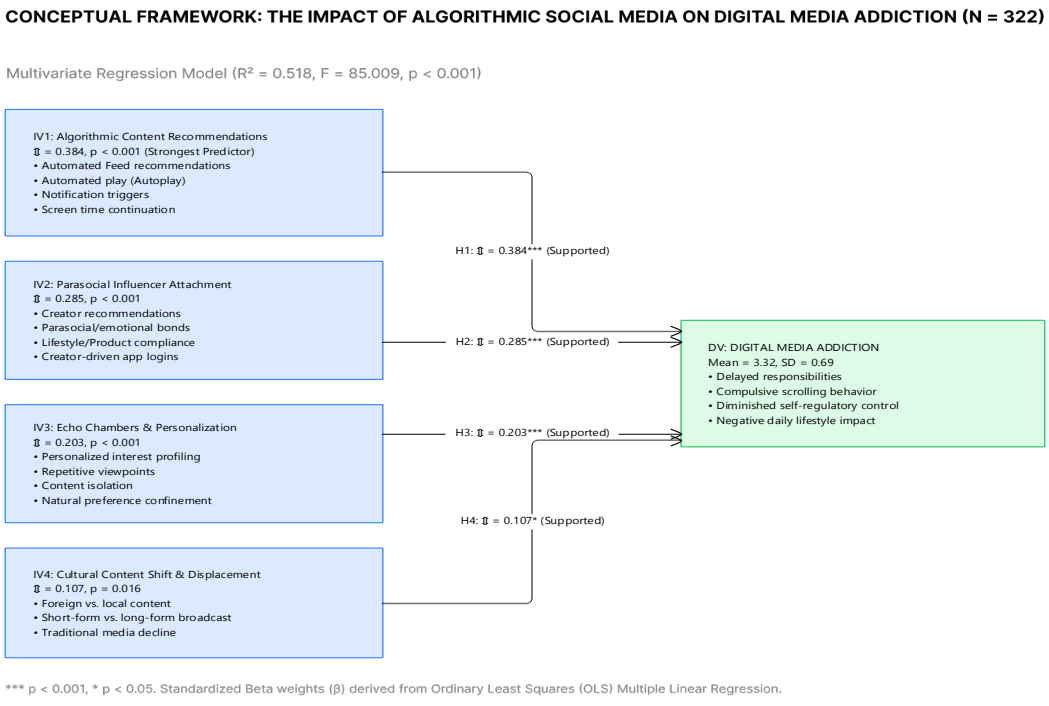
The theory provides for four independent variables that stand for a unique sociotechnological aspect and are related to one dependent variable.

Independent Variables (IVs)

- IV1 – Algorithmic Content: This refers to recommendations, auto-play, and notifications
- IV2 – Influencer Bonding: This includes parasocial relationships as well as the creation of content through algorithms
- IV3 – Echo Chamber Effect: This relates to personal feedback loops and restricting oneself from different views
- IV4 – Cultural Shift: It refers to the consumption of foreign, fast content rather than traditional broadcasting

Dependent Variable (DV)

- Digital Media Addiction (DV): Measured based on reduced daily focus, addiction to scrolling, loss of control, and neglecting responsibilities.



1.5 Hypotheses Formulation

- Hypothesis 1: Content recommendations via algorithm have a positive impact on digital media addiction.
- Hypothesis 2: Parasocial interactions with celebrities on digital platforms have a statistically significant positive impact on digital media addiction.
- Hypothesis 3: Echo chamber and personalization algorithms have a statistically significant positive impact on digital media addiction.
- Hypothesis 4: Cultural content shifts via algorithm have a positive impact on digital media addiction.

Chapter 2: Literature Review and Theoretical Framework

2.1 Introduction to Recommendation Engine Architectures

Modern social media platforms are no longer just software applications but intelligent machine learning-based environments. One of the most crucial aspects of social media platforms like TikTok and YouTube is the recommendation system, which involves two modules: candidate generation and ranking. For identifying similarities, collaborative filtering can be used between users or media objects. After this list of videos has been made, this multi-task ranking algorithm will rank these videos based on the click rates, watching time, and probability of engagement. The reason for this is that these algorithms operate in real time and optimize the newsfeed so that the right piece of content is shown to the user before he or she leaves the application.

2.2 Psychological Frameworks: Skinner's Operant Conditioning

Behavioral capture in today's UI/UX is based on behavioral psychology, which involves operant conditioning theory by B.F. Skinner. He proved that behaviors could be reinforced best through the use of variable ratio schedule. From an engineering perspective, there are certain elements in software design that behave in the same way as digital slot machines. With the refresh of the algorithm-driven feed by the user, there is no guarantee that the following video is going to be highly interesting or boring. This factor makes the process more unpredictable, hence increases the amount of dopamine produced by the reward pathway in the brain. Because of this unpredictability of the result, humans have biological tendencies to keep scrolling the application for the next positive feedback.

2.3 Media System Dependency (MSD) Theory

As per Media Systems Dependency Theory proposed by Sandra Ball-Rokeach and Melvin DeFleur, the more an individual depends upon the media to meet his/her needs, the greater importance this media holds in his/her life. Today's world is the age of algorithms where digital media forms the means of communication, information sharing, and entertainment. In governing the crucial channels,

the algorithms create a situation of dependence. People who try to disassociate themselves face cognitive resistance, fear of missing out, and feeling lonely, all of which drive them back into the algorithm's world.

2.4 Collaborative Filtering and Echo Chamber Confinement

The "filter bubble" or "echo chamber effect" is one of the main results of algorithms generated through machine learning. It happens because such algorithms are designed to show users information in accordance with their beliefs and previous actions. This will make the system better for the user in terms of their experience and use. It will however trap the user in a closed feedback loop that has been customized according to his/her personal needs. This will limit the chances of the user coming across new information.

2.5 Parasocial Interaction within the Creator Economy

The parasocial interaction theory explains why people tend to develop a one-sided and emotional attachment towards the characters appearing in media. The recommendation system makes this possible by presenting niche video producers or influencers whose profile characteristics match those of the viewer psychologically. As the user watches hundreds of micro-videos by the same influencer, an impression of friendliness and closeness develops. With such a parasocial relationship, the user's approach towards these media changes from that of entertainment to duty due to the "bond" developed.

Chapter 3: Research Methodology

3.1 Instrument Design and Operational Scale

The research methodology involved the development of a quantitative questionnaire for the collection of empirical data that is related to the research goals. The questionnaire had six major parts and consisted of 30 questions.

SURVEY INSTRUMENT ARCHITECTURAL MAPPING

Section	Target Dimension	Item Count	Measurement Scale
Section 1	Demographic Profile	5 Items	Categorical / Nominal
Section 2	Algorithmic Content (IV1)	5 Items	5-Point Frequency (1 to 5)
Section 3	Influencer Attach. (IV2)	5 Items	5-Point Likert Agreement (1 to 5)
Section 4	Echo Chambers (IV3)	5 Items	5-Point Likert Agreement (1 to 5)
Section 5	Cultural Shift (IV4)	5 Items	5-Point Likert Agreement (1 to 5)
Section 6	Screen Time & DV Addiction	5 Items	Ratio, Binary & 5-Point Likert

3.1.1 Questionnaire Structure

1. Section 1: General Information (Items 1.1-1.5) - Participant demographic attributes were identified, such as age group, gender, occupation, language preference for content consumption, and digital media platform of choice.

2. Section 2: Algorithm-driven Content Recommendation (IV1_Algo, Items 2.6-2.10) - Measured participants' behavioral exposure to algorithmic features offered by the platform, including recommendation feed, video recommendations, autoplay, push notifications, and the continued passive browsing.

3. Section 3: Parasocial Relationship with Influencers (IV2_Influencer, Items 3.11-3.15) - Evaluated participants' connection with algorithm-driven influencers, the frequency of visits to the application motivated by influencer stalking, parasocial closeness, and purchases made due to the lifestyle of an influencer.

4. Section 4: Echo Chamber Effect and Personalization (IV3_Echo, Items 4.16-4.20): Perceived algorithmic understanding, repetition of views, resistance to learning about content outside known likes, perceived loss of control, and confirmation bias were measured.

5. Section 5: Shifts in Cultural Content (IV4_Culture, Items 5.21-5.25): Tests focused on the mix of local (Sri Lankan) versus foreign content, exposure to foreign cultural media, reduced watching of broadcasting TV, and liking micro-content (such as TikTok/Reels).

6. Section 6: Screen Time and Addiction Markers (Items 6.26-6.30): A numerical variable that indicated the precise number of hours used daily (Item 6.26); an indicator variable reflecting pre-sleep use, which was defined as use close to bedtime or within 30 minutes of getting up (Item 6.27); and three Likert type items measuring procrastination, subjective addiction, and its effect on lifestyle (DV_Addiction, Items 6.28-6.30).

3.1.2 Scaling and Quantification Rationale

For the execution of the Pearson correlation and OLS regression tests, sections 2 to 5 as well as the indicators of addiction were defined by the following Likert scales symmetric about a range from 1 to 5:

- Frequency scale (section 2): 1 = never, 2 = rarely, 3 = sometimes, 4 = often, 5 = always.
- Agreement scale (sections 3-6): 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree.

The standardization of Likert scale numbers into an interval from 1 to 5 allowed us to give the same quantitative meaning to all scale items and calculate the constructs' composite scores by arithmetic average in IBM SPSS Statistics.

$$FactorScore = \frac{1}{k}\sum_{i=1}^{k} X_i$$

In this case, k is the number of variables that are going to be allocated to the target variable, whereby k = 5 for independent variables, while k = 3 for dependent addiction variable.

3.1.3 Pilot Testing and Validity Assessment

Before the actual surveys were sent out, a pilot survey was carried out on 15 computer science students. The pilot survey considered the following aspects:

- Face and Content Validity: Every survey question was checked to ensure that each question matched the relevant construct.
- Interface Compatibility: This involved testing the survey interface for compatibility on the tablet, mobile, and desktop devices.
- Survey Completion Time: It was ensured that the survey could be completed in 4 to 6 minutes to prevent respondent fatigue.

From the above test, it was observed that the data provided by the respondents was unambiguous, and all the data collected was properly entered in the intake database.

3.2 Target Population and Eligibility Criteria

3.2.1 Target Population

The target audience for this research was the digital media users aged between 18 and 40 years living in Sri Lanka. The selection of the aforementioned age group is based on the following premises:

- First, digital nativity: Persons belonging to this group have the maximum number of smartphones and use mobile broadband speedily along with being engaged with short videos streaming apps frequently.
- Second, occupation/education sensitiveness: Both students and workers belonging to the age group face various problems related to productivity that make them prone to mental exhaustion, staying awake at nights, and interfering with their studies/work.

3.2.2 Eligibility Criteria

- Inclusion criteria:

1. Age of participants should be between 18 to 40 years at the time of conducting the study.
2. Should use at least one algorithm-based video application (TikTok, YouTube, Instagram, Facebook).
3. Sri Lankan nationals with the ability to complete the questionnaire in English.

- Exclusion criteria:

1. Participants whose age is below 18 years or more than 40 years.
2. Not using any algorithm-based social networking sites.
3. Incomplete responses or computer-generated responses with no valid authentication.

3.3 Sampling Strategy and Administration

3.3.1 Sampling Methodology

Considering the infeasibility of developing a sampling frame to encompass all the social media users within Sri Lanka, non-probabilistic sampling was adopted along with the method of digital snowball

sampling. In order to overcome any demographic biases that might occur within the sample distribution, the following fields were considered:

- Academics (public/governmental universities and private universities such as the students' associations at NIBM)
- Business (software development companies and other businesses)
- General public (technology-related websites, media groups, and regionally relevant social media groups)

3.3.2 Survey Administration Protocol

Data collection was done through the use of Google Forms. This website link together with a brief description of the academic focus of the study was shared via multiple channels on the internet such as WhatsApp groups, LinkedIn professional networks, and university notice boards.

Participants were told about their rights before they took part in the survey:

- Participation was completely voluntary and no financial or organizational pressure was exerted on them.
- Participants' identities remained anonymous and their names, phone numbers, and exact addresses were not noted.
- Information was collected purely for academic purposes.

3.4 Data Integrity, Screening, and Sample Size ($\boldsymbol{N = 322}$)

3.4.1 Sample Size Determination

According to the requirements set out in the specifications of the coursework, the required minimum number of respondents was N = 250. The aim was to obtain enough statistical power for multivariate regression analysis ($1 - \beta \geq 0.80$ at $\alpha = 0.05$). Considering the possible drop-out during the data cleaning process, data collection continued up to N = 322 respondents in the final sample. This is an increase of 72 people or 28.8%.

3.4.2 Ingestion Controls and Integrity Verification

The steps taken to achieve rigorous data hygiene while collecting data include the following:

1) Google OAuth One Response Per Token Rule: The implementation of Google OAuth meant that participants had to submit one response per authentication token to prevent multiple submissions.
2) Mandatory Question Rule: All thirty questions in the survey instrument had to be answered by the participants, thus leaving no missing data (0.0% missing data, N=322).

3.4.3 Data Preprocessing and Transformation Pipeline

After completing the data collection process, the data matrix containing unstructured data was exported to an .xlsx file, which was then imported to IBM SPSS Statistics 27 (SC_CW_analyse.sav) to manipulate the data, as explained below:

1. Variable Labeling: The long survey questions were renamed to concise variable labels (Q1.1 to Q6.30).
2. Variable Recoding: The age group string variable (1.1: What is your age group?) was recoded into a numeric variable (Age_Numeric) using AUTORECODE (1 = 18-24, 2 = 25-30, 3 = 31-40) to make it amenable for parametric testing via One-Way ANOVA.
3. Scale Development: The Likert scale variables were summed up using SPSS COMPUTE command with MEAN() to develop the continuous scales: IV1_Algo, IV2_Influencer, IV3_Echo, IV4_Culture, and DV_Addiction.

Chapter 4: Results and Statistical Analysis

4.1 Descriptive Statistics of Evaluated Constructs

A descriptive statistical analysis was performed on all N = 322 observations to understand the central tendencies, dispersion and limits of distributional bounds of the four independent variables and the primary dependent construct. All the constructs scores were computed using the arithmetic mean of their constituent 5-point Likert scale items and bound the observed factor scores between 1.00 and 5.00.

Table 4.1 Descriptive Statistics for Studies Constructs (N = 322)

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
IV1_Algo	322	1.40	4.80	3.3248	.71839
IV2_Influencer	322	1.60	5.00	3.5224	.76712
IV3_Echo	322	1.00	5.00	3.0553	.73424
IV4_Culture	322	1.40	4.80	3.2758	.71858
DV_Addiction	322	1.00	5.00	3.3261	.69285
Valid N (listwise)	322				

The descriptive statistics in Table 4.1 suggest high to moderate levels of participation for all studies:

- The study constructed Parasocial Influencer Attachment (IV2_Influencer) had the highest overall agreement (M = 3.5224, SD = 0.76712), indicating that participants have an affinity toward algorithmically suggested digital creators, and that this social media social presence is a pervasive part of users' social media consumption.

- The study constructed Digital Media Addiction (DV_Addiction) yielded a composite mean of 3.3261 (SD = 0.69285) which indicates that participants experience moderate to severe disruption of their schedules and perceived addiction to the algorithmic feeds.

- Respondents from Algo mean average was 3.3248 ($SD = 0.71839$). This shows that most of them interacted with automatic playlists and suggested queues.

- Respondents from IV4_Culture mean average was 3.2758 ($SD = 0.71858$). This shows that they reported the displace ment of Sri Lankan media (both conventional and OTT) by foreign media.

- Respondents from IV3_Echo mean average was 3.0553 ($SD = 0.73424$). This shows that users perceived the recommendation systems as clearly viewpoint constrictive. The focus group discussions showed the presence of different levels of awareness for user segments regarding the recommendation systems.

4.2 Demographic and Platform Utilization Analysis

The baseline data set captured five demographic and behavioral characteristics in the sample (N = 322) as follows:

Table 4.2: Demographic Characteristics of the Sample (N = 322)

Statistics

		1.1: What is your age group?	1.2: What is your gender?	1.3: What is your current occupation?	1.4: What language do you prefer most for digital content?	1.5: Which digital media platform do you spend the most time on?
N	Valid	322	322	322	322	322
	Missing	0	0	0	0	0

1.1: What is your age group?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-24	186	57.8	57.8	57.8
	25-30	124	38.5	38.5	96.3
	31-40	12	3.7	3.7	100.0
	Total	322	100.0	100.0	

1.2: What is your gender?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	159	49.4	49.4	49.4
	Male	152	47.2	47.2	96.6

	Prefer not to say	11	3.4	3.4	100.0
	Total	322	100.0	100.0	

1.3: What is your current occupation?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Employed	122	37.9	37.9	37.9
	Self-Employed	44	13.7	13.7	51.6
	Student	126	39.1	39.1	90.7
	Unemployed	30	9.3	9.3	100.0
	Total	322	100.0	100.0	

1.4: What language do you prefer most for digital content?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	English	59	18.3	18.3	18.3
	Mix	121	37.6	37.6	55.9
	Sinhala	116	36.0	36.0	91.9
	Tamil	26	8.1	8.1	100.0
	Total	322	100.0	100.0	

1.5: Which digital media platform do you spend the most time on?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Facebook	21	6.5	6.5	6.5
	Instagram	74	23.0	23.0	29.5
	TikTok	124	38.5	38.5	68.0
	YouTube	103	32.0	32.0	100.0

Total	322	100.0	100.0	

Behavioral trends from demographic profiling include:

- Concentration of the Target Population: The sample is composed of 57.76% digital natives aged between 18 and 24 years and 38.51% of the 25 to 30 years young working professionals. This gives an appropriate case for empirical testing to examine the extent of mobile algorithmic engagement.

- MICRO BROADCASTING: TikTok is the dominant platform with 38.51% of respondents citing it as their most utilized platform (38.51%). YouTube (31.99%) and Instagram (22.98%) follow respectively. Traditional broadcast social channels (i.e. Facebook) represent only 6.52% of respondents' primary daily usage.
- Bedtime and Morning Usage Telemetry (6.27): 56.52% of respondents (n = 182) admitted to using social media within 30 minutes of either going to sleep or waking up, indicating a very high level of attachment to social media and an adjacent behavior to sleep cycle.

- Continuous Daily Media Engagement (6.26): The daily enter train time for entertainment was reported to be 4.57 hours (SD = 1.50, range 0.50-8.60), indicating very heavy and prolonged engagement with digital content and feeds.

4.3 Hypothesis Testing: Chi-Square Test of Independence

A Pearson Chi-Square Test of Independence was performed to evaluate if there is a significant relationship between user Gender (Item 1.2) and User's Primary Digital Media Platform (Item 1.5)

Table 4.3: Gender vs. Primary Digital Media Platforms Crosstabulation

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
1.2: What is your gender? * 1.5: Which digital media platform do you spend the most time on?	322	100.0%	0	0.0%	322	100.0%

1.2: What is your gender? * 1.5: Which digital media platform do you spend the most time on? Crosstabulation

Count

		1.5: Which digital media platform do you spend the most time on?		
		Facebook	Instagram	TikTok
1.2: What is your gender?	Female	16	43	51
	Male	5	30	68
	Prefer not to say	0	1	5
Total		21	74	124

1.2: What is your gender? * 1.5: Which digital media platform do you spend the most time on? Crosstabulation

Count

		1.5: Which digital media platform do you spend the most time on?	
		YouTube	Total
1.2: What is your gender?	Female	49	159
	Male	49	152
	Prefer not to say	5	11
Total		103	322

Table 4.4: Chi-Square Test Output

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	13.044[a]	6	.042
Likelihood Ratio	14.068	6	.029
N of Valid Cases	322		

a. Expected counts for 4 (33.3%) cells are less than 5. The minimum expected count is 0.72.

Statistical Interpretation:

The Pearson Chi-Square value is statistically significant at the α = 0.05 level.

$$\chi^2(6, N = 322) = 13.044, p = 0.042 (p < 0.05)$$

The asymptotic significance value of 0.042 is less than the standard critical alpha of 0.05, meaning the null hypothesis of independence can be rejected. There is a statistically significant relationship between a user's gender and their preferred digital media platform.

Male users preferred TikTok (n = 68) over female users (n = 51). Female users preferred Instagram (n = 43) over male users (n = 30) and preferred Facebook (n = 16) over male users (n = 5). There was a preference for YouTube (n = 49) by both male and female users.

(Note on Methods: Four cells produced expected frequencies below 5 due to the low sample of users preferring "not to say" (n = 11), however the Likelihood Ratio test (χ^2 = 14.068, p = 0.029) indicates statistical stability of the relationship).

4.4. Hypothesis Testing: One-Way ANOVA

To investigate whether there are differences in the levels of digital media addiction across age brackets, a One-Way ANOVA was conducted. The age string variable was converted to numeric using the AUTORECODE procedure to create the variable Age_Numeric (1 = 18 to 24, 2 = 25 to 30, 3 = 31 to 40) and was analyzed against the continuous dependent variable DV_Addiction.

Table 4.5. Group Level Digital Media Addiction by Age Cohorts

Descriptives

DV_Addiction

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum
					Lower Bound	Upper Bound	
18-24	186	3.3889	.68788	.05044	3.2894	3.4884	1.00
25-30	124	3.2124	.69497	.06241	3.0888	3.3359	1.33
31-40	12	3.5278	.62697	.18099	3.1294	3.9261	2.33
Total	322	3.3261	.69285	.03861	3.2501	3.4020	1.00

Descriptives

DV_Addiction

	Maximum
18-24	5.00
25-30	4.67
31-40	4.67
Total	5.00

Table 4.6. One-Way ANOVA Results (DV_Addiction by Age_Numeric)

ANOVA

DV_Addiction

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	2.825	2	1.413	2.979	.052
Within Groups	151.269	319	.474		
Total	154.094	321			

Statistical Interpretation

The one-way ANOVA examined the variance across different age groups against the within-groups residual variance, which was measured to be statistically significant ($p < 0.05$).

$$F(2,319) = 2.979, p = 0.052$$

Using the standard 95% confidence interval (α=0.05), the test yields a marginally non-significant result (p=0.052>0.05). Thus, the null hypothesis is not rejected at the α=0.05 level.

As such, minor differences do exist whereby individuals in the 25 to 30 age bracket reported lower addiction scores (M=3.2124) compared to those in the 18 to 24 (M=3.3889) and 31 to 40 (M=3.5278) age brackets. However, the differences in the levels of addiction across the various adult age brackets were found to be statistically insignificant. Algorithmic digital media addiction is more pervasive across all adult age groups (brackets) in the study compared to remaining in a particular age group (bracket).

4.5 Bivariate Relationship: Pearson Correlation Analysis

To analyze the strength, direction, and significance of the bivariate associations between each independent variable and digital media addiction, a two-tailed Pearson correlation analysis was conducted on the full sample (N = 322).

Table 4.7 Pearson Correlation Matrix (N = 322)

Correlations

		IV1_Algo	IV2_Influencer	IV3_Echo	IV4_Culture
IV1_Algo	Pearson Correlation	1	.273**	.440**	.318**
	Sig. (2-tailed)		.000	.000	.000
	N	322	322	322	322
IV2_Influencer	Pearson Correlation	.273**	1	.388**	.314**
	Sig. (2-tailed)	.000		.000	.000
	N	322	322	322	322
IV3_Echo	Pearson Correlation	.440**	.388**	1	.422**
	Sig. (2-tailed)	.000	.000		.000
	N	322	322	322	322
IV4_Culture	Pearson Correlation	.318**	.314**	.422**	1
	Sig. (2-tailed)	.000	.000	.000	
	N	322	322	322	322
DV_Addiction	Pearson Correlation	.585**	.502**	.527**	.404**
	Sig. (2-tailed)	.000	.000	.000	.000
	N	322	322	322	322

Correlations

		DV_Addiction
IV1_Algo	Pearson Correlation	.585**
	Sig. (2-tailed)	.000
	N	322
IV2_Influencer	Pearson Correlation	.502**
	Sig. (2-tailed)	.000
	N	322
IV3_Echo	Pearson Correlation	.527**
	Sig. (2-tailed)	.000
	N	322
IV4_Culture	Pearson Correlation	.404**
	Sig. (2-tailed)	.000
	N	322
DV_Addiction	Pearson Correlation	1
	Sig. (2-tailed)	
	N	322

. Correlation is significant at the 0.01 level (2-tailed).

Statistical Interpretation of Bivariate Relationships

The correlation matrix shows that the four independent variables have positively linear relationships with Digital Media Addiction at the 99% confidence level ($p < 0.001$) as follows:

Algorithmic Content Recommendations (IV1_Algo): Displays the highest positive correlation with digital media addiction ($r = 0.585$, $p < 0.001$). Increased user interaction with automated content (e.g., autoplay and suggested content) and triggered (e.g. push) content (i.e. feed) further intensifies their sense of addictive behavior.

Echo Chambers and Personalization (IV3_Echo): Shows the second highest correlation with addiction ($r = 0.527$, $p < 0.001$). In this case, the reinforced viewpoint (perspective) and content (ies) isolation further retains the user within the application (or platform).

Parasocial Influencer Attachment (IV2_Influencer): Shows a positive correlation with addiction ($r = 0.502$, $p < 0.001$). In the case of digital media, emotional attachment to social media influencers can enhance a user's sense of addiction.

Cultural Content Shift (IV4_Culture): Displays a positive correlation with addiction ($r = 0.404$, $p < 0.001$). In this context, high-velocity, short-form, foreign social content can increase a user's sense of digital media addiction.

Additionally, the independent variables also show moderate levels of inter-correlation (ranging from $r = 0.273$ to $r = 0.440$), therefore, the independent constructs do not show collinearity with redundancy ($r < 0.70$).

4.6 Predictive Modeling: Multiple Linear Regression Analysis

An OLS Multiple Linear Regression was conducted employing the ENTER method in order to determine whether the four independent constructs predict DV_Addiction concurrently.

Table 4.8: Model Summary

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.719[a]	.518	.511	.48428

a. Predictors: (Constant), IV4_Culture, IV2_Influencer, IV1_Algo, IV3_Echo

Table 4.9: ANOVA (Overall Model Significance)

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	79.748	4	19.937	85.009	.000[b]
	Residual	74.346	317	.235		
	Total	154.094	321			

a. Dependent Variable: DV_Addiction

b. Predictors: (Constant), IV4_Culture, IV2_Influencer, IV1_Algo, IV3_Echo

Table 4.10: Regression Coefficients

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	.267	.173		1.541	.124
	IV1_Algo	.370	.043	.384	8.673	.000
	IV2_Influencer	.258	.039	.285	6.599	.000
	IV3_Echo	.191	.045	.203	4.264	.000
	IV4_Culture	.103	.043	.107	2.416	.016

a. Dependent Variable: DV_Addiction

Statistical Interpretation of the Regression Results

1. Goodness of Fit and Variance Explained

The regression analysis resulted in a correlation coefficient R=0.719 which demonstrates a strong multivariate relationship between the predictor set and the dependent construct. Coefficient of determination:

$$R^2 = 0.518 (AdjustedR^2 = 0.511)$$

Thus, \$51.8\%\$ of the total variance of Digital Media Addiction is explained by the set of four independent constructs. The rest \$48.2\%\$ of variance can be explained by the presence of other psychological, environmental or neurobiological factors not included in the model.

2. Significance of the Regression Model

ANOVA test of the regression model produced the omnibus test statistic of:

$$F(4,317) = 85.009, p < 0.001$$

As the p-value is well below the 0.001 level, the regression equation significantly predicts the dependent construct.

3. Specification of the Regression Equation

By using the unstandardized regression coefficients (B), the empirical linear regression equation is as follows:

$$\hat{Y}_{Addiction} = 0.267 + 0.370(X_{IV1_Algo}) + 0.258(X_{IV2_Influencer}) + 0.191(X_{IV3_Echo}) + 0.103(X_{IV4_Culture})$$

4. Evaluation of Individual Predictors and Relative Weights

All four independent variables were statistically significant ($p < 0.05$) in the multivariate analysis.

- Algorithmic Content Recommendations (IV1_Algo): Was found to be the strongest predictor ($\beta = 0.384$, $t = 8.673$, $p < 0.001$). With other factors being the same, an increase in the number of algorithmic recommendations is associated with an increase in addiction by 0.370.
- Parasocial Influencer Attachment (IV2_Influencer): Was found to be the second strongest predictor ($\beta = 0.285$, $t = 6.599$, $p < 0.001$). Increase in parasocial attachment to influencers is associated with an increase in addiction by 0.258.
- Echo Chambers and Personalization (IV3_Echo): Was also found to be a significant predictor ($\beta = 0.203$, $t = 4.264$, $p < 0.001$). More echo chambers are associated with an increase in addiction by 0.191.
- Cultural Content Shift (IV4_Culture): Was found to be a positive predictor ($\beta = 0.107$, $t = 2.416$, $p = 0.016$). More cultural displacement is associated with an increase in addiction by 0.103.

Table 4.11 Hypotheses Testing Summary Results

Hypothesis	Structural Path	Standardized Beta (β)	t-value	Sig. (p)	Empirical Verdict
$\boldsymbol{H_1}$	Algorithmic Content → Digital Addiction	.384	8.673	.000 ($p < .001$)	Supported
$\boldsymbol{H_2}$	Influencer Attachment → Digital Addiction	.285	6.599	.000 ($p < .001$)	Supported
$\boldsymbol{H_3}$	Echo Chambers → Digital Addiction	.203	4.264	.000 ($p < .001$)	Supported
$\boldsymbol{H_4}$	Cultural Content Shift → Digital Addiction	.107	2.416	.016 ($p < .05$)	Supported

Chapter 5: Discussion, Implications and Recommendations

5.1 Discussion and Synthesis of Empirical Findings

Findings from this study show that various algorithms used in social media platforms promote digital media addiction among young adults. Results from the multivariate linear regression showed that it explained 51.8% of the variance in digital media addiction ($R^2 = 0.518$, Adjusted $R^2 = 0.511$, $F(4,317) = 85.009$, $p < 0.001$). All the four independent variables were found to be statistically significant and therefore supported Hypotheses H1, H2, H3, and H4.

Summary of Regression Model Parameters and Outcomes

Hypothesis	**Structural Relationship**	**Beta (β)**	**t-value**	**p-value**	**Status**
H1	Algorithmic Content → Digital Addiction	0.384	8.673	< 0.001	Supported

H2	Influencer Attachment → Addiction	0.285	6.599	< 0.001	Supported
H3	Echo Chambers → Digital Addiction	0.203	4.264	< 0.001	Supported
H4	Cultural Content Shift → Addiction	0.107	2.416	0.016	Supported

Model Summary

Parameter	**Value**
Dependent Variable	DV_Addiction
R^2	0.518
F-statistic	F(4, 317) = 85.009
Model Significance	$p < 0.001$

5.1.1 Prevalence of Algorithms' Content Mechanisms (H1)

Out of the four independent variables, Algorithmic Content Recommendations (IV1_Algo) was the most dominant single predictor in the regression model ($\beta = 0.384$, $t = 8.673$, $p < 0.001$). It also had the strongest bivariate correlation with digital media addiction ($r = 0.585$, $p < 0.001$). From the empirical data, dominance of mechanism of behavioral capture driven by design thinking and use of algorithmic recommendation systems, integrated with technologies such as zero-latency scrolling, autoplay, background pre-fetching, and push notifications, was observed. Traditional intentional goal-directed browsing has been replaced with recommendation systems where users are passively consumption content. The recommendation systems use variable-ratio reinforcement where every time a user refreshes their feed, they are rewarded with content they are not expecting which releases dopamine. Ultimately, the design thinking and the integration of Algorithmic recommendation systems within social media create an operant conditioning and behavioral capture system among social media users.

5.1.2 The Significance of Parasocial Influencer Attachment (H_2)

Parasocial Influencer Attachment (IV2_Influencer) turned out to be the second most important predictor ($\beta = 0.285$, $t = 6.599$, $p < 0.001$). Moreover, it had the highest level of respondents' agreement ($M = 3.5224$, $SD = 0.76712$).

This means that recommendation systems do not upload videos randomly but analyze the content produced by the creators and promote those who fit the psychological profile of the users. These content creators influence the formation of perceptions of one-sided relationships (parasocial relationships) each time the user consumes any type of content on the platform. As a result, social validation and emotional attachment develop in such a way that the platform turns into the main source of social activity.

5.1.3 The Influence of Echo Chambers and Personalization (H_3)

Echo Chambers and Personalization (IV3_Echo) proved to have a significant positive relationship with digital media addiction ($\beta = 0.203$, $t = 4.264$, $p < 0.001$) as well as high bivariate correlation ($r = 0.527$, $p < 0.001$).

The mean score of this independent variable was the lowest ($M = 3.0553$, $SD = 0.73424$). This helps increase the users' comfort level and strengthen their ideological beliefs through a feedback loop. Algorithms predict the users' preferences, confirm and promote the content that supports the users'

ideas, which leads to confirmation bias. As a result, the content becomes less mentally challenging for the user and allows him/her to spend more time online.

5.1.5 Synthesis of Demographic and Behavioral Results

Demographic results provide vital information about behavioral patterns associated with the use of platforms:

- Gender-Platform Association: According to the Pearson Chi-Square test, there was a statistically significant relationship between gender and the choice of the digital platform used ($\chi^2(6, N=322) = 13.044$, $p = 0.042$). Male respondents were characterized by a clear predominance of the preference for TikTok (n = 68) over females (n = 51); females were more active users of Instagram (n = 43 vs. n = 30 among males) and Facebook (n = 16 vs. n = 5 among males). YouTube was equally popular among men (n = 49) and women (n = 49).

- Age Ubiquity: As the One-Way ANOVA revealed, there was no significant difference between the mean addiction scores among adult age groups at the 95% confidence interval ($F(2, 319) = 2.979$, $p = 0.052$). Although individuals aged 25–30 had a slightly lower mean addiction score ($M = 3.2124$) compared to those aged 18–24 ($M = 3.3889$) and 31–40 ($M = 3.5278$), the p-value proves that algorithmic addiction affects young adults of all age groups included in the survey instead of the youngest one.

- Sleep and Daily Life Interference: Based on descriptive results, 56.52% (n = 182) of the respondents admitted to the use of social media within 30 minutes after getting up and/or before going to sleep, which means their average daily screen time devoted to leisure is 4.57 hours (SD = 1.50). This indicates that algorithmic addiction directly interferes with the biological sleep process and work.

5.2 Practical Implications for the Industry and Computing Professionals

5.2.1 Organizational Efficiency and Focus of the Workforce

The participants, aged 18 to 25, have a mean daily use of 4.57 hours of digital entertainment. The participants also have extreme cases of nomophobia, with 56.52% saying they use digital feeds as soon as they wake up or as they go to bed. The employees of this survey (37.89%) and the self-employed participants (13.66%) can also be at risk of cognitive disassociation due to constant work notifications. Notification pop-ups and sound alerts can also lead to context-switching and disrupt deep-work which is essential for software development and other cognitive tasks.

Additionally, digital media's effects on bedtime scrolling is detrimental to the productivity of the employees due to insomnia, reduced cognitive processing speed, working memory and Emotional Regulation which can lead to emotional exhaustion and burnout at work.

5.2.2 Software Engineering and UI/UX Design

From a computing perspective, these results represent that designers and developers of digital media have achieved their goal of adding digital media to our daily lives and integrate it to a degree that will cause behavioral and psychological effects to the users. Manipulative features such as autoplay and notifications are designed to intentionally exploit and monetize human attention. These features encourage its users to develop a habit and psychological dependency to the product which is detrimental to the users' cognitive and emotional well-being. The features of digital media are optimized by product teams to ultimately lead to regression and a loss of user autonomy.

5.3 Architectural and Policy Recommendations

To address the negative impact of algorithmic dependency, this paper will present a set of recommendations in the form of a mitigation framework with multiple layers.

Multi-Tiered Digital Addiction Mitigation Framework

Level	Intervention Focus	Primary Mechanism
Architecture	Ethical Interface Design	Natural friction, opt-in autoplay, limits
Organizational	Workplace Well-Being	Asynchronous norms, notification blackouts
Individual	Behavioral Self-Control	Cache auditing, grayscale, time boundaries

5.3.1 Platform and Software Engineering Recommendations

1. Platform developers should introduce ethical friction by breaking zero-friction consumption patterns. Designers can implement pagination and/or feeds in a batch, as opposed to infinite, scroll. They can also implement thoughtful interfaces to create pauses, for example, using a mandatory confirmation dialog after 30 minutes of uninterrupted playback.

2. Client-side media players should not have autoplay enabled by default. If designers wanted to leverage autoplay, they should implement user actions, and thus deliberate control, to initiate the next automatic consumption of media.

3. Interfaces should allow control over recommendation systems. In addition, users should have the ability to adjust the exploration-versus-exploitation balance and view why videos are recommended to them.

4. Rate-limit budgets should be implemented by mobile and desktop clients of operating systems to batch social media notifications.

5.3.2 Organizational and Institutional Recommendations

1. Residential and corporate clients should adopt asynchronous communication.
2. Organizations should create policies that prevent work communications outside of office hours.
3. Organizations should implement digital wellbeing programs.

5.3.3 Recommendations for Individual Behavioral Change

1. Mobile Interface Sanitization: Users can change the display on their devices to grayscale at night and turn off alerts and notifications for applications they do not need.

2. Cache Management for Algorithms: Users can reinitialize advertising identification numbers, clear their browser cache and application cache to reduce personalization and the creation of user-specific echo chambers.

3. Physical Barriers to Devices: Individuals should put the charging station for their devices in another room, preferably the bedroom, to not use the device at night.

5.4Limitations of the Study and Future Research

4.4.1 Limitations of the Study

- Sampling: Non-probability convenience sampling and a digital snowball method was used for sampling in Sri Lanka. While a sample size of N=322 offered good statistical power, the sample tended to be university-based students (39.13%) and young employed professionals (37.89%), and as such, is not representative of the rural and older populations.

- Cross-Sectional Design: With a cross-sectional design, relationships can only be described at a single point in time. Therefore, in the absence of repeated measurements of the study variables, the regression model was unable to describe true causal relationships.
- Subjective Self-Reporting Bias: Surveys relied on participants to estimate their own screen time and addiction, which may result in various biases, e.g. recall bias, social desirability bias, as compared to objective data from participants' devices.

5.4.2 Future Research Directions

- Objective Telemetry: Future research in the field of computing and Human Computer Interaction (HCI) should employ objective telemetry concerning device usage via the Android Screen Time or iOS Screen Time API, followed by empirical self-reporting studies to validate the results.

- Longitudinal Interventional Studies: Future research should perform longitudinal studies to examine the effects of changes in behavior and cognition over time using user data gathered by open-source screen-time interceptors.

- Comparative Algorithmic Auditing: Future research should perform black-box auditing to compare different recommendation systems and architectures in various social media and content distribution platforms (e.g., TikTok's sequence-modeling recommendation systems compared to YouTube's candidate generation pipelines) in order to analyze specific components of algorithms that contribute to the formation of compulsion reinforcement.

The findings of this study support the empirical model and show that for people aged between 18 to 25 years, digital media automation and recommendation pipelines (β=0.384) and parasocial relationships (β=0.285) have the most significant impact. Echo chambers and cultural shift (β = 0.177) play an auxiliary role. Overall, the results show that the design of such applications causes the development of consumers' addiction and not their personal control loss (R^2=0.518). The responsibility of computing professionals lies in thinking outside the box of recommendation systems and other architectures that maximize the engagement and developing ethical and friction-aware systems that will protect users from consumption and compulsion to interact with digital media.

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